

Discrete Structures (Discrete Mathematics)

Predicates and
Quantifiers

Bitwise Operations

- Computer represents information using bits. A bit is a symbol with two possible values, namely, 0 (zero) and 1 (one).
- 1 represents T (true) and 0 represents F (false).
- A variable is called a Boolean Variable if its value is either true or false.
- A Boolean Variable can be represented by a bit.
- A **bit string** is a series of Boolean values. Length of the string is the number of bits.
 - 101010011 is nine Boolean values in one string

Bitwise Operations

- We can then do operations on these Boolean strings
 - Each column is its own Boolean operation

Operations	Operator	Bit String1	Bit String 2	Result
Bitwise XOR	$\oplus$	0101 1010	1011 0100	1110 1110
Bitwise OR	$\vee$	0101 1010	1011 0100	1111 1110
Bitwise AND	$\wedge$	0101 1010	1011 0100	00010000

Why Predicate Logic?

- **Propositional Logic is not expressive enough**
 - It cannot adequately express the meaning of statements in mathematics and in natural language

Example 1:

“Every computer connected to the university network is functioning properly.”

- No rules of propositional logic allow us to conclude the truth of the statement.

Why Predicate Logic?

Example 2:

- “There is a computer on the university network that is under attack by an intruder.”

**Predicate Logic is more expressive
and powerful**

Predicate Logic

- Proposition, YES or NO?

- $3 + 2 = 5$ 

- $X + 2 = 5$ 

- $X + 2 = 5$ for any choice of X in $\{1, 2, 3\}$ 

- $X + 4 = 5$ for some X in $\{1, 2, 3\}$ 

- Computer X is under attack by an intruder 

Propositional Functions(Example)

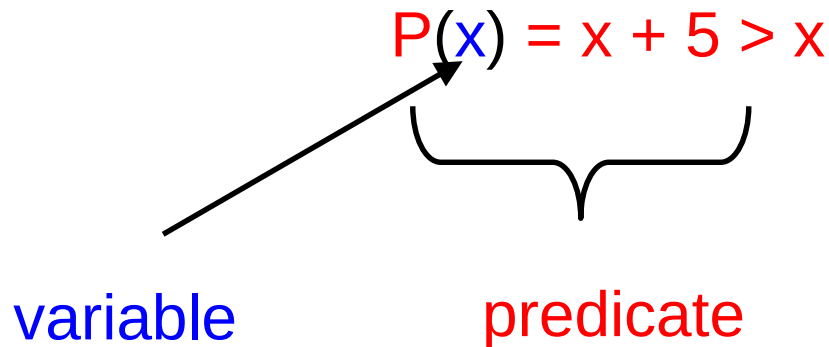
- “x is greater than 3” or $(x > 3)$
 - The variable x: subject of the statement
 - “is greater than 3”: predicate
 - $P(x)$: propositional function P at x
- Let $P(x) = x > 3$
 - $P(x)$ has no truth values (x is not given a value)
 - $P(10)$ is true: The proposition $10 > 3$ is true.
 - $P(1)$ is false: The proposition $1 > 3$ is false.
 - $P(x)$ will create a proposition when given a value

Propositional Functions(Example)

- Let $A(x)$ = “Computer x is under attack by an intruder.”
- Suppose computers on campus, only CS2 and MATH1 are currently under attack by intruders.
- What are truth values of $A(\text{CS1})$, $A(\text{CS2})$, and $A(\text{MATH1})$?
 - The statement $A(\text{CS1})$ by setting $x = \text{CS1}$ in the statement “Computer x is under attack by an intruder.”
 - CS1 is not on the list of computers currently under attack, $A(\text{CS1})$ is false.
 - CS2 and MATH1 are on the list of computers under attack, $A(\text{CS2})$ and $A(\text{MATH1})$ are true.

Propositional Functions

- Functions with multiple variables:
 - $P(x,y) = x + y == 0$
 - $P(1,2)$ is false, $P(1,-1)$ is true
 - $P(x,y,z) = x + y == z$
 - $P(3,4,5)$ is false, $P(1,2,3)$ is true
 - $P(x_1, x_2, x_3 \dots x_n) = \dots$
- Anatomy of a propositional function



Predicates

- A predicate is a declarative statement (Propositional function) with at least one variable (i.e. unknown value).
- A predicate, or propositional function, is a function that takes some variable(s) as arguments and returns True or False.

Predicates

- Suppose $Q(x,y) = "x > y"$

Proposition, YES or NO?

$Q(x,y)$

No

$Q(3,4)$

Yes

$Q(x,9)$

No

Predicate, YES or NO?

$Q(x,y)$

Yes

$Q(3,4)$

No

$Q(x,9)$

Yes

Quantification

- Quantification expresses the extent to which a predicate is true over a range of elements.
- It is process of specifying the scope or range of variables within a proposition.
- In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications.
- The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

Types of Quantifiers

- A quantifier is “an operator that limits the variables of a proposition”.
- Two types:
 - Universal quantifier
 - Existential quantifier

Universal Quantifiers

- Represented by an upside-down A: $\forall$
 - It means “for all”
 - Let $P(x) = x+1 > x$
- We can state the following:
 - $\forall x P(x)$
 - English translation: “for all values of x , $P(x)$ is true”
 - English translation: “for all values of x , $x+1 > x$ is true”

Besides “for all”, universal quantification can be expressed in many other ways: “for every”, “all of”, “for each”, “given any”, “for arbitrary”, “for each” and “for any”

Universal Quantifiers

Statement holds for all elements in a given set (Universe)

For example, $\forall x (P(x))$ asserts that the predicate P is true for every element x in the given domain.

Universal Quantifiers

- You need to specify the **universe of quantification**!
 - What values x can represent
 - Called the “domain of discourse” or “universe of discourse”
 - Or just “domain” or “universe”
- *The meaning of the universal quantification of $P(x)$ changes when we change the domain. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not defined.*

Universal Quantifiers

- Let the universe of discourse be the real numbers.
- Let $P(x) = x/2 < x$
 - Not true for the negative numbers!
 - Thus, $\forall x P(x)$ is false, When the domain is all the real numbers
- In order to prove that a universal quantification is true, it must be shown for **ALL** cases
- In order to prove that a universal quantification is false, it must be shown to be false for **only ONE** case

Universal Quantifiers

- Let $P(x)$ is “ $x^2 > 0$.” To show that the statement $\forall x P(x)$ is false where the universe of discourse consists of all integers, we give a counterexample.
- $x = 0$ is a counterexample because $x^2 = 0$ when $x = 0$, so that x^2 is not greater than 0 when $x = 0$.

Universal Quantification

- Given some propositional function $P(x)$ And values in the universe $x_1 \dots x_n$
- The universal quantification $\forall x P(x)$ implies:
 - $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$

Question

- What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement $x^2 < 10$ and the domain consists of the positive integers not exceeding 4?

Question

- What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement $x^2 < 10$ and the domain consists of the positive integers not exceeding 4?

Solution:

- The statement $\forall x P(x)$ is the same as the conjunction
$$P(1) \wedge P(2) \wedge P(3) \wedge P(4),$$
- Because $P(4) \equiv 4^2 < 10$, is **false**, it follows that $\forall x P(x)$ is **false**.

Existential Quantification

- Represented by an backwards E: $\exists$
 - It means “there exists”, “there is”, “for some”, etc.
 - Let $P(x) = x+1 > x$
- We can state the following:
 - $\exists x P(x)$
 - English translation: “there exists (a value of) x such that $P(x)$ is true”
 - English translation: “for at least one value of x , $x+1 > x$ is true”
 - English translation: “for some x , $P(x)$ ”

Existential Quantification

- The existential quantifier of $P(x)$ is the proposition:
 - “ $P(x)$ is true for some x in the universe of discourse.”
- $\exists x P(x)$ is TRUE if there is an x for which $P(x)$ is true.
- $\exists x P(x)$ is FALSE if $P(x)$ is false for every single x .

Existential Quantification

- Note that you still have to specify your universe
- Let $P(x) = x+1 < x$
 - There is no numerical value x for which $x+1 < x$
 - Thus, $\exists x P(x)$ is false

Existential Quantification

- Let $P(x) = x+1 > x$
 - There is a numerical value for which $x+1 > x$
 - In fact, it's true for all of the values of x . Thus, $\exists x$ $P(x)$ is true
- In order to show an existential quantification is **true**, you only have to **find ONE value**
- In order to show an existential quantification is **false**, you have to show **it's false for ALL values**

Existential Quantification

- **Example:** Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists xP(x)$, where the domain consists of all real numbers?
- **Solution:** Because “ $x > 3$ ” is sometimes true—for instance, when $x = 4$ the existential quantification of $P(x)$, which is $\exists xP(x)$, is true.

Existential Quantification

- **Example:** Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?
- **Solution:** Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists x Q(x)$, is false.

Existential Quantification

- Given some propositional function $P(x)$ And values in the universe $x_1 \dots x_n$
- The existential quantification $\exists x P(x)$ implies:
 - $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$

Summary

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.
- e.g $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$.

Binding Variables

- When a quantifier is used on a variable x , we say that this occurrence of the variable is **bound**.
- An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be **free**.
- The part of a logical expression to which a quantifier is applied is called the **scope** of the quantifier.
- All the variables that occur in a logical expression must be bound or set equal to a particular value to turn into a proposition.

Binding Variables

- Examples:
- $P(x)$ x is free
- $P(5)$ x is bound to 5
- $\forall x P(x)$ x is bound by quantifier

Binding Variables

- $\exists x (P(x) \wedge Q(x)) \vee (\forall x R(x))$
 - All variables are bound.
- The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$ because $\exists x$ is applied only to $P(x) \wedge Q(x)$, and not to the rest of the statement.
- Similarly, the scope of the second quantifier, $\forall x$, is the expression $R(x)$.
- That is, the existential quantifier binds the variable x in $P(x) \wedge Q(x)$ and the universal quantifier $\forall x$ binds the variable x in $R(x)$.

Binding Variables

- $\exists x (x + y = 1)$
 - x is bound by $\exists x$ and y is free; thus not a proposition
- $(\exists x P(x)) \vee Q(x)$
 - The x in $Q(x)$ is not bound; thus not a proposition
- $(\exists x P(x)) \vee (\forall x Q(x))$
 - Both x values are bound; thus it is a proposition
- $\exists x (P(x) \wedge Q(x)) \vee (\forall y R(y))$
 - All variables are bound; thus it is a proposition
- $(\exists x P(x) \wedge Q(y)) \vee (\forall y R(y))$
 - The y in $Q(y)$ is not bound; thus not a proposition

A note on quantifiers

- Recall that $P(x)$ is a propositional function
 - Let $P(x)$ be “ $x == 0$ ”
- Recall that a proposition is a statement that is either true or false
 - $P(x)$ is not a proposition
- There are two ways to make a propositional function into a proposition:
 - Supply it with a value
 - For example, $P(5)$ is false, $P(0)$ is true
 - Provide a quantification
 - For example, $\forall x P(x)$ is false and $\exists x P(x)$ is true
 - Let the universe of discourse be the real numbers

Translating From English to Logical Expressions

- Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.

- **Solution:**

- Assume domain is students in the class

“For every student in this class, that student has studied calculus.”

“For every student x in this class, x has studied calculus.”

$C(x)$ = “ x has studied calculus.”

$$\forall x C(x)$$

Negating Quantified Expressions

- Consider the statement
“Every student in this class has studied calculus.”
- This statement is a universal quantification, namely, $\forall x C(x)$,
 - $C(x)$ is the statement “ x has studied calculus”
 - Domain consists of the students in the class.
- The negation of this statement is
 - “It is not the case that every student in this class has studied calculus.”
 - This is equivalent to “There is a student in this class who has not studied calculus.”
- This is simply the existential quantification of the negation of the original propositional function, namely, $\exists x \neg C(x)$.

Negating Quantified Expressions

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

Negation	Equivalent Statement	When is Negation True?	When False?
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Negating Quantified Expressions

- **Example:**

What are the negations of the statements “There is an honest politician” and “All Americans eat cheeseburgers”?

Solution:

- Let $H(x)$ denote “ x is honest.” Then the statement “There is an honest politician” is represented by $\exists x H(x)$, where the domain consists of all politicians.
- The negation of this statement is $\neg \exists x H(x)$, which is equivalent to $\forall x \neg H(x)$. This negation can be expressed as “Every politician is dishonest.” or “Not all politicians are honest.”

Negating Quantified Expressions

Solution:

- $C(x)$ denote “ x eats cheeseburgers.” Then the statement “All Americans eat cheeseburgers” is represented by $\forall x C(x)$, where the domain consists of all Americans.
- The negation of this statement is $\neg \forall x C(x)$, which is equivalent to $\exists x \neg C(x)$. This negation can be expressed as “Some American does not eat cheeseburgers” and “There is an American who does not eat cheeseburgers.”

Negating Quantified Expressions

- What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = x)$?

De Morgan's Laws for Quantifiers

$$\begin{aligned} \bullet \neg \exists x P(x) &= \neg(P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)) \\ &\equiv \neg P(x_1) \wedge \neg P(x_2) \wedge \dots \wedge \neg P(x_n) \\ &\equiv \forall x \neg P(x) \end{aligned}$$

$$\begin{aligned} \bullet \neg \forall x P(x) &= \neg(P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)) \\ &\equiv \neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n) \\ &\equiv \exists x \neg P(x) \end{aligned}$$

Translating From English to Logical Expressions

- Let $C(x)$ = “ x has studied calculus.”
 $S(x)$ = “person x is student in this class.”
 The domain for x consists of all people.
- “For every person x , if person x is a student in this class then x has studied calculus.”
- The statement can be expressed as $\forall x(S(x) \rightarrow C(x))$.

Translating From English to Logical Expressions

- Let $P(x)$ = “ x can speak Russian”

$Q(x)$ = “ x knows the computer language C++.”

Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives.

The domain for quantifiers consists of all students at your school.

- There is a student at your school who can speak Russian and who knows C++.
- There is a student at your school who can speak Russian but who doesn't know C++.

Translating From English to Logical Expressions

- Every student at your school either can speak Russian or knows C++.
- No student at your school can speak Russian or knows C++.

The Four Aristotelian Forms

1. All A's are B's
 2. Some A's are B's
 3. No A's are B's
 4. Some A's are not B's
- These are four of the most common quantificational sentences used in quantificational reasoning.

The First Aristotelian Form

- The Form: *All A's are B's*
- Example: All comedian are funny.
 - Rephrase: For every x , if x is a comedian then x is funny
 - Translation: $\forall x (\text{Comedian}(x) \rightarrow \text{Funny}(x))$
 - This translation has the form: $\forall x (A(x) \rightarrow B(x))$
- General Fact
 - All A's are B's translates as $\forall x (A(x) \rightarrow B(x))$

The Second Aristotelian Form

- The Form: *Some A's are B's*
- Example: Some comedian are funny
 - Rephrase: Some thing x is both comedian and funny
 - Translation: $\exists x (\text{Comedian}(x) \wedge \text{Funny}(x))$
 - This translation has the form: $\exists x (A(x) \wedge B(x))$
- General Fact
 - Some A's are B's translates as $\exists x (A(x) \wedge B(x))$

The Second Aristotelian Form

- We've learned two facts:
 1. All A's are B's translates as $\forall x (A(x) \rightarrow B(x))$
 2. Some A's are B's translates as $\exists x (A(x) \wedge B(x))$

The Third Aristotelian Form

- The Form: *No A's are B's*
- Example: No students are failed
 - Rephrase: For every x, if x is a student then x is not failed
 - Translation: $\forall x (\text{Student}(x) \rightarrow \neg \text{Failed}(x))$
 - This translation has the form: $\forall x (A(x) \rightarrow \neg B(x))$
- General Fact
 - No A's are B's translates as $\forall x (A(x) \rightarrow \neg B(x))$

The Fourth Aristotelian Form

- The Form: *Some A's are not B's*
- Example: Some excuses are not believable
 - Rephrase: For some x, x is an excuse and x is not believable
 - Translation: $\exists x (\text{Excuse}(x) \wedge \neg \text{Believable}(x))$
 - This translation has the form: $\exists x (A(x) \wedge \neg B(x))$
- General Fact
 - Some A's are not B's translates as $\exists x (A(x) \wedge \neg B(x))$

Summary

- The Aristotelian Forms and Their Translations

- All A's are B's $\forall x (A(x) \rightarrow B(x))$
- Some A's are B's $\exists x (A(x) \wedge B(x))$
- No A's are B's $\forall x (A(x) \rightarrow \neg B(x))$
- Some A's are not B's $\exists x (A(x) \wedge \neg B(x))$

Predicates - Examples

$L(x)$ = “x is a lion.”

$F(x)$ = “x is fierce.”

$C(x)$ = “x drinks coffee.”

Universe of discourse
is all creatures

$$\forall x (L(x) \rightarrow F(x))$$

All lions are fierce.

Some lions don't drink coffee.

$$\exists x (L(x) \wedge \neg C(x))$$

Some fierce creatures don't drink coffee.

$$\exists x (F(x) \wedge \neg C(x))$$

Predicates - Examples

$B(x)$ = “x is a hummingbird.”

$L(x)$ = “x is a large bird.”

$H(x)$ = “x lives on honey.”

$R(x)$ = “x is richly colored.”

Universe of discourse
is all creatures

All hummingbirds are richly colored.

$$\forall x (B(x) \rightarrow R(x))$$

No large birds live on honey.

$$\forall x (L(x) \rightarrow \neg H(x))$$

Birds that do not live on honey are dully colored.

$$\forall x (\neg H(x) \rightarrow \neg R(x))$$

Example

- Use predicates and quantifiers to express the system specifications “Every mail message larger than one megabyte will be compressed.”
- Where the variable m has the domain of all mail messages and the variable y is a positive real number
- Let $S(m, y)$ = “Mail message m is larger than y megabytes”
 $C(m)$ = “Mail message m will be compressed.”
- Then the specification “Every mail message larger than one megabyte will be compressed” can be represented as
$$\forall m(S(m, 1) \rightarrow C(m)).$$

Example

- Use predicates and quantifiers to express the system “If a user is active, at least one network link will be available.”
- Let $A(u)$ = “User u is active” where the variable u has the domain of all users.
- Let $S(n, x)$ = “Network link n is in state x ,” where n has the domain of all network links and x has the domain of all possible states for a network link.
- Then the specification “If a user is active, at least one network link will be available” can be represented by
$$\exists u A(u) \rightarrow \exists n S(n, \text{available}).$$

Example

- Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English.
 - a) Some old dogs can learn new tricks.
 - b) No rabbit knows calculus.
 - c) Every bird can fly.
 - d) There is no dog that can talk.
 - e) There is no one in this class who knows French and Russian.

Chapter Reading

- **Chapter 1**, Kenneth H. Rosen, Discrete Mathematics and Its Applications, Section 1.4

Exercise of Section 1.3 (For Practice) – Page 53

- Question # 1, 2, 5, 6, 7, 8, 10, 11, 12, 14, 17, 18, 35, 36, 59(a, b, c), 60(a, b, c), 61(a, b, c, d)